# Quizrr Chapter-Wise Test for JEE Advanced - 2025

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**Q.1**

$g(n)=\int_0^{n^2+n+1} e^{x/2-[x/2]}\left(\frac{x}{2}-\left[\frac{x}{2}\right]\right) d(x-[x]); n \in N$ then $g(n)$

Here [.] represents GIF.

**(A)** has minimum value as $\frac{1}{4}+\sqrt{\mathrm{e}}$

**(B)** has maximum value as $3-\sqrt{e}$

**(C)** has minimum value as $\frac{3}{4}-\sqrt{\frac{e}{4}}$

**(D)** none of these

**Q.2**

$\int_0^{\infty} \mathrm{x}^{2\mathrm{n}+1} \cdot \mathrm{e}^{-\mathrm{x}^2} \mathrm{dx}$ is equal to $(\mathrm{n} \in \mathrm{N})$

**(A)** $\frac{\mathrm{n}!}{2}$

**(B)** $\mathrm{n}!$

**(C)** $(\mathrm{n}-1)!$

**(D)** $\frac{\mathrm{n}}{2}$

**Q.3**

$\int\limits_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$ is

**(A)** $\pi^2/4$

**(B)** $\pi^2$

**(C)** $0$

**(D)** $\pi/2$

**Q.4**

Let $I_1=\int_0^1 \frac{e^x dx}{1+x}$ and $I_2=\int_0^1 \frac{x^2 dx}{e^{x^3}(2-x^3)}$ then $\frac{I_1}{I_2}$ is equal to

**(A)** $3/\mathrm{e}$

**(B)** e/ 3

**(C)** $3\mathrm{e}$

**(D)** $1/3\mathrm{e}$

**Q.5**

The value of $\int_0^{\pi} \frac{x^2 \sin^4 x}{\pi^2+2x^2-2x\pi} dx=$

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**(A)** $\frac{\pi}{16}$

**(B)** $\frac{3\pi}{8}$

**(C)** $\frac{\pi}{8}$

**(D)** $\frac{3\pi}{16}$

**Q.6**

The value of $\int \left(\frac{\sec^2 x - 7}{\sin^7 x}\right) dx$ is

**(A)** $\frac{\tan x}{(\sin x)^7} + C$

**(B)** $\frac{\tan^7 x}{\sin x} + C$

**(C)** $\left(\frac{\tan x}{\sin x}\right)^7 + C$

**(D)** none of these

**Q.7**

Evaluate: $\int \frac{x^2}{(x \sin x + \cos x)^2} dx$

**(A)** $\frac{-\mathrm{x} \sec \mathrm{x}}{\mathrm{x} \sin \mathrm{x} + \cos \mathrm{x}} + \tan \mathrm{x} + \mathrm{C}$

**(B)** $\frac{-x \sec x}{x \sin x + \cos x} - \tan x + c$

**(C)** $\frac{\mathrm{x} \sec \mathrm{x}}{\mathrm{x} \sin \mathrm{x} + \cos \mathrm{x}} + \tan \mathrm{x} + \mathrm{c}$

**(D)** None of these

**Q.8**

$\int [\sin(\log x) + \cos(\log x)] dx$ is equal to

**(A)** $\sin(\log x) + \cos(\log x) + C$

**(B)** $x \sin(\log x) + C$

**(C)** $x \cos(\log x) + C$

**(D)** None of these

**Q.9**

If $I_n = \int_0^\infty e^{-x} (\sin x)^n dx (n \in N > 1)$ then $-$

**(A)** $I_n = \frac{n(n-1)}{n^2+1} I_{n-2}$

**(B)** $I_n = \frac{n(n-1)}{(n-1)^2+1} I_{n-1}$

**(C)** $101 \frac{\mathrm{I}_{10}}{\mathrm{I}_8} = 90$

**(D)** $82 \frac{\mathrm{I}_{10}}{\mathrm{I}_9} = 90$

**Q.10**

$\int \sqrt{\frac{1-x}{1+x}} dx$ is equal to

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**(A)** $\sin^{-1} x+\sqrt{1-x^2}+C$

**(B)** $-\cos^{-1} x+\sqrt{1-x^2}+C$

**(C)** $\sin^{-1} \mathrm{x}-2 \sqrt{1-\mathrm{x}^2}+\mathrm{C}$

**(D)** $2 \sin^{-1} x-\sqrt{1-x^2}+C$

**Q.11**

Let $\mathrm{f}(\mathrm{x})=\mathrm{x}+\int_{-1}^{1}\left(\mathrm{x}-\mathrm{tx}^2\right) \mathrm{f}(\mathrm{t}) \mathrm{dt}$. Then which of the following alternative (s) is/are correct-

**(A)** $\lim_{x \to 0} f(x)=2$

**(B)** $\mathrm{f}'\left(\frac{-1}{3}\right)=1$

**(C)** $\mathrm{f}$ is continuous and derivable on $\mathrm{R}$

**(D)** maximum value of $\mathrm{f}(\mathrm{x})$ does not exist

**Q.12**

**Passage**

Let $A=\int_0^1 \frac{e^x}{x+1} dx$ then answer the following questions in terms of $A$.

**Question:**

$\int_0^1 \frac{x^2 e^x}{x+1} dx$ equals -

**(A)** $\mathrm{A}-\mathrm{e}$

**(B)** $e-2+A$

**(C)** $2+\mathrm{A}$

**(D)** $2-\mathrm{e}+\mathrm{A}$

**Q.13**

**Passage**

Let $A=\int_0^1 \frac{e^x}{x+1} dx$ then answer the following questions in terms of $A$

**Question:**

$\int_0^1 \left(\frac{x}{x+1}\right)^2 e^x dx$ equals -

**(A)** $A-\frac{e}{2}$

**(B)** $\frac{\mathrm{e}}{2}+1-\mathrm{A}$

**(C)** $\frac{\mathrm{e}}{2}-\mathrm{A}$

**(D)** $A-1+\frac{e}{2}$

**Q.14**

**Passage**

Let $A=\int_0^1 \frac{e^x}{x+1} dx$ then answer the following questions in terms of $A$

**Question:**

$\int_0^1 \frac{x^3}{(x+1)^3} e^x dx$ equals -

**(A)** $\frac{\mathrm{e}}{2}-2+\mathrm{A}$

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**(B)** $3A - \frac{3e}{2}$

**(C)** $3 - \frac{A}{2} - \frac{9e}{8}$

**(D)** None of these

**Q.15**

**Passage**

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$

**Question:**

If $\mathrm{I} = \mathrm{k}\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{xdx}$ then $\mathrm{k}$ is equal to -

**(A)** 1

**(B)** 5

**(C)** 10

**(D)** 20

**Q.16**

**Passage**

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$

**Question:**

If $\mathrm{I} = \mathrm{c}\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 2\mathrm{xdx}$ then c equals -

**(A)** 5

**(B)** 10

**(C)** 20

**(D)** $5/2$

**Q.17**

**Passage**

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$

**Question:**

The value of I equals -

**(A)** $\frac{5\pi}{4}$

**(B)** $\frac{5\pi}{16}$

**(C)** $\frac{5\pi}{32}$

**(D)** $\frac{5\pi}{8}$

**Q.18**

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Match the column -

| Column I | Column II |
|---|---|
| (A) If $F(x) = \int \frac{x+\sin x}{1+\cos x}dx$ and $F(0) = 0$, then values equal to and greater than $F(\pi/2) =$ | (p) $-\pi/2$ |
| (B) Let $F(x) = \int e^{\sin^{-1}x}\left(1-\frac{x}{\sqrt{1-x^2}}\right)dx$ and $F(0) = 1$, if $F(1/2) = \frac{k\sqrt{3}e^{\pi/6}}{\pi}$ then values equal to and greater than $k =$ | (q) $\pi/3$ |
| (C) Let $F(x) = \int \frac{dx}{(x^2+1)(x^2+9)}$ and $F(0) = 0$, if $F(\sqrt{3}) = \frac{5}{36}k$, then values equal to and greater than $k =$ | (r) $\pi/4$ |
| (D) Let $F(x) = \int \frac{\sqrt{\tan x}}{\sin x \cos x}dx$ and $F(0) = 0$ if $F(\pi/4) = \frac{2k}{\pi}$ then $k =$ | (s) $\pi$ |

**(A)** A-s ; B-s ; C-q,r,s ; D-s

**(B)** A-s ; B-s,q ; C-r ; D-p

**(C)** A ; B-s ; C-q,r,s ; D-q

**(D)** A-p,r ; B-s,q ; C-r,s,p ; D-s

**Q.19**

If $\int_0^{2\pi} \ln\left(\sec^2 x\right) dx = 2k^2 \sin^{-1}\left(\frac{k}{2}\right) \ln k$, then find the value of $[k]$, (where [ . ] denotes greatest integer function).

**Q.20**

If the value of $\int_0^{100\pi} \left(\left[\cot^{-1} x\right] + \left[\tan^{-1} x\right]\right) dx$ is $100\pi + P\cot(P)$, then the value of $P$ is?
(where [ . ] denotes greatest integer function).

## Answers & Solutions

**Q.1 Answer:**

none of these

**Solution:**

$$g(n) = \int_0^{n^2+n+1} e^{\{x/2\}} \left\{\frac{x}{2}\right\} d\{x\}$$

$$= \left(n^2+n+1\right) \int_0^1 e^{\{x/2\}} \left\{\frac{x}{2}\right\} dx$$

$$= \left(n^2+n+1\right) \int_0^1 e^{x/2} \left(\frac{x}{2}\right) dx$$

$$= \left(n^2+n+1\right)(2-\sqrt{e})$$

**So, minimum value is** $6-3\sqrt{e}$

**Q.2 Answer:**

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$\frac{n!}{2}$

**Solution:**

$I = \int_0^\infty x^{2n+1} \cdot e^{-x^2} dx$

$= \int_0^\infty x^{2n} . x . e^{-x^2} dx$

**Put** $x^2 = t \Rightarrow xdx = \frac{dt}{2}$

$I = \frac{1}{2} \int_0^\infty t^n . e^{-t} dt = \frac{1}{2} \left[ -t^n \cdot e^{-t} \Big|_0^\infty + n \int_0^\infty t^{n-1} . e^{-t} dt \right]$

$= \frac{1}{2} \left[ 0 + n \int_0^\infty t^{n-1} . e^{-t} dt \right] = \frac{n}{2} \Gamma(n-1)$

**Hence,** $I = \frac{n!}{2}$

**Q.3 Answer:**

$\pi^2$

**Solution:**

$2 \int\limits_{-\pi}^{\pi} \frac{x}{1+\cos^2 x} dx + 2 \int\limits_{-\pi}^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$

$= 0 + 2 \int\limits_{-\pi}^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$

$= 2 \times 2 \int\limits_{0}^{\pi} \frac{x \sin x}{1+\cos^2 x}$

$= 4 \int\limits_{0}^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$

$= 4 \times \frac{\pi}{2} \int\limits_{0}^{\pi} \frac{\sin x}{1+\cos^2 x} dx$

$\left( \text{by using } \int\limits_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int\limits_0^\pi f(\sin x) dx \right)$

$= 4 \frac{\pi}{2} \times 2 \times \int\limits_{0}^{\pi/2} \frac{\sin x}{1+\cos} dx$

$= 4\pi \left( \tan^{-1} \cos x \right)_{1}^{0}$

**(By putting** $\cos x = t$**)**

$= 4\pi \times \left( \frac{\pi}{4} - 0 \right)$

$= \pi^2$

**Q.4 Answer:**

$3e$

**Solution:**

$I_1 = \int_0^1 \frac{e^x dx}{1+x}, I_2 = \int_0^1 \frac{x^2 dx}{e^{x^3}(2-x^3)}$

**In** $I_2$**, put** $1 - x^3 = t$

$$\Rightarrow I_2 = \frac{1}{3} \int_1^0 \frac{-dt}{e^{1-t}(1+t)}$$

$$= \frac{1}{3e} \int_0^1 \frac{e^t dt}{1+t} = \frac{1}{3e} I_1$$

$$\Rightarrow \frac{I_1}{I_2} = 3e$$

**Q.5 Answer:**

$\frac{3\pi}{16}$

**Solution:**

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$I = \int_0^\pi \frac{x^2 \sin^4 x dx}{\pi^2+2x^2-2x\pi} = \int_0^\pi \frac{x^2 \sin^4 x dx}{x^2+(\pi-x)^2} \longrightarrow (1)$

$I = \int_0^\pi \frac{(\pi-x)^2 \sin^4 x dx}{(\pi-x)^2+x^2} \longrightarrow$ **(2)**

**Now, (1) + (2) we get,**

$$2I = \int_0^\pi \sin^4 x dx = \int_0^\pi \left(\sin^2 x\right)^2 dx$$

$$\therefore 2I = \int_0^\pi \left(\frac{1-\cos 2x}{2}\right)^2 dx$$

$$\therefore 2I = \frac{1}{4}\int_0^\pi \left(1 - 2\cos 2x + \cos^2 2x\right) dx$$

$\therefore 2I = \frac{1}{4}\left[(x)_0^\pi - (\sin 2x)_0^\pi + \frac{1}{2}\left(x + \frac{\sin 4x}{4}\right)_0^\pi\right]$

$2I = \frac{1}{4}\left[\pi - 0 + \frac{1}{2}(\pi)\right]$

$I = \frac{1}{8} \cdot \frac{3\pi}{2}$

$\therefore I = \frac{3\pi}{16}$

**Q.6 Answer:**

$\frac{\tan x}{(\sin x)^7} + C$

**Solution:**

$\int \left(\frac{\sec^2 x}{\sin^7 x} - \frac{7}{\sin^7 x}\right) dx$

$= \int \frac{\sec^2 x}{\sin^7 x} dx - \int \frac{7}{\sin^7 x} dx = I_1 - I_2$

**Now,** $I_1 = \int \frac{\sec^2 x}{\sin^7 x} dx = \frac{\tan x}{\sin^7 x} + 7\int \frac{\tan x \cdot \cos x}{\sin^8 x} dx$

$= \frac{\tan x}{\sin^7 x} + I_2$

$\therefore \quad I_1 - I_2 = \frac{\tan x}{\sin^2 x} + C$

**Q.7 Answer:**

$\frac{-\mathrm{x}\sec \mathrm{x}}{\mathrm{x}\sin \mathrm{x}+\cos \mathrm{x}} + \tan \mathrm{x} + \mathrm{C}$

**Solution:**

**Let** $I = \int \frac{x^2}{(x\sin x+\cos x)^2} dx$

**Multiplying and dividing it by** $(x\cos x)$**, we get**

$I = \int (x\sec x) \cdot \frac{(x\cos x)}{(x\sin x+\cos x)^2} dx$

$I = x\sec x \cdot \int \frac{x\cos x}{(x\sin x+\cos x)^2} dx - \int \left\{\frac{d}{dx}(x\sec x)\right\} \left\{\int \frac{x\cos x}{(x\sin x+\cos x)^2} dx\right\} dx$

$= x\sec x \cdot \frac{-1}{(x\sin x+\cos x)} dx - \int (x\sec x \cdot \tan x + \sec x) \cdot \frac{-1}{(x\sin x+\cos x)} dx$

$= \frac{-x\sec x}{(x\sin x+\cos x)} + \int \frac{(x\sin x+\cos x)}{\cos^2 x \cdot (x\sin x+\cos x)} dx$

$= \frac{-x\sec x}{(x\sin x+\cos x)} + \int \sec^2 x dx$

$= \frac{-x\sec x}{(x\sin x+\cos x)} + \tan x + C$

**Q.8 Answer:**

$x\sin(\log x) + C$

**Solution:**

**Put** $\log \mathrm{x} = \mathrm{z} \Rightarrow \mathrm{x} = \mathrm{e}^{\mathrm{z}} \Rightarrow \mathrm{dx} = \mathrm{e}^{\mathrm{z}}\mathrm{dz}.$

$\therefore \quad \int [\sin(\log x) + \cos(\log x)] dx$

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$= \int e^z(\sin z + \cos z)dz$
$= e^z \sin z + C = x\sin(\log x) + C$

**Q.9 Answer:**

$I_n = \frac{n(n-1)}{n^2+1} I_{n-2}$

**Solution:**

$I_n = -(\sin x)^n e^{-x}\big|_0^\infty + \int_0^\infty n(\sin x)^{n-1} \cdot \cos x e^{-x} dx$
$= -n(\sin x)^{n-1}\cos x e^{-x}\Big|_0^\infty + \int_0^\infty \left(n(n-1)(\sin x)^{n-2}\cos^2 x - n(\sin x)^{n-1}\sin x\right) e^{-x}dx$
$I_n = \int_0^\infty n(n-1)\left((\sin x)^{n-2} - \sin^n x\right) - n\sin^n x)\right) e^{-x}dx$
$I_n = \int_0^\infty n(n-1)\left((\sin x)^{n-2} - \sin^n x\right) - n\sin^n x)\right) e^{-x}dx = \frac{n(n-1)}{n^2+1} I_{n-2}$
$\frac{I_{10}}{I_8} = \frac{10\times 9}{101} = \frac{90}{101}$

**Q.10 Answer:**

$\sin^{-1} x + \sqrt{1-x^2} + C$

**Solution:**

$\int \sqrt{\frac{1-x}{1+x}} dx$
$= \int \left(\frac{1}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}\right) dx$
$= \sin^{-1} x + \sqrt{1-x^2} + C = -\cos^{-1} x + \sqrt{1-x^2} + C$
$= \sin^{-1} x + \sqrt{1-x^2} + C$

**Q.11 Answer:**

$f'\left(\frac{-1}{3}\right) = 1$

**Solution:**

**We have,** $f(x) = x + x\int_{-1}^1 f(t)dt - x^2\int_{-1}^1 tf(t)dt$
$f(x) = (1+A)x - Bx^2$
**where** $A = \int_{-1}^1 f(t)dt$ **and** $B = \int_{-1}^1 tf(t)dt$
**Now,** $A = \int_{-1}^1 f(t)dt = \int_{-1}^1 \left((1+A)t - Bt^2\right) dt$
$\Rightarrow A = -2B\int_0^1 t^2 dt \quad \therefore \quad A = \frac{-2B}{3}$
**Similarly,** $B = \int_{-1}^1 tf(t)dt = \int_{-1}^1 t\left((1+A)t - Bt^2\right) dt$
$\Rightarrow B = 2(A+1)\int_0^1 t^2 dt \quad \therefore B = \frac{2}{3}(A+1)$
**On solving (2) and (3), we get** $A = \frac{-4}{13}, B = \frac{6}{13}$
$\therefore$ **From eq. (1), we get** $f(x) = \frac{9}{13}x - \frac{6}{13}x^2 \Rightarrow f'(x) = \frac{9}{13} - \frac{12}{13}x$
$\therefore \quad f'\left(-\frac{1}{3}\right) = \frac{9}{13} - \frac{12}{13}\left(-\frac{1}{3}\right) = \frac{9}{13} + \frac{4}{13} = \frac{13}{13} = 1$

**Q.12 Answer:**

$2 - e + A$

**Solution:**

**(i)** $\int_0^1 \frac{x^2 e^x}{x+1} dx = \int_0^1 \frac{\left(x^2-1+1\right)e^x}{x+1} dx$
$= \int_0^1 (x-1)e^x dx + \int_0^1 \frac{e^x}{x+1} dx$
$= \left((x-1)e^x - e^x\right)_0^1 + A = ((-e) - (-1-1) + A = 2 - e + A$

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**Q.13 Answer:**

$\frac{e}{2} - A$

**Solution:**

(ii) $\int_0^1 \left(\frac{x}{x+1}\right)^2 e^x dx = \int_0^1 \left(\frac{(x+1)^2-(2x+1)}{(x+1)^2}\right) e^x dx$

$I = \int_0^1 e^x dx - 2\int_0^1 \frac{e^x}{x+1} dx + \int_0^1 \frac{1}{(x+1)^2} e^x dx$

$I = (e-1) - 2A + \left(-\frac{e}{2} + 1 + A\right) = \frac{e}{2} - A$

$\int_0^1 \frac{x^3}{(x+1)^3} e^x dx = \int_0^1 \frac{x^3+1-1}{(x+1)^3} e^x dx$

$= \int_0^1 \frac{(x+1)^3-3x(x+1)-1}{(x+1)^3} e^x dx$

$= 1 - 3\int_0^1 \frac{x+1-1}{(x+1)^2} e^x dx - 1\int_0^1 \frac{1}{(x+1)^3} e^x dx$

**Q.14 Answer:**

$3 - \frac{A}{2} - \frac{9e}{8}$

**Solution:**

(iii) $\int_0^1 \frac{x^3}{(x+1)^3} e^x dx = \int_0^1 \frac{x^3+1-1}{(x+1)^3} e^x dx = \int_0^1 \frac{(x+1)^3-3x(x+1)-1}{(x+1)^3} e^x dx$

$= 1 - 3\int_0^1 \frac{x+1-1}{(x+1)^2} e^x dx - 1\int_0^1 \frac{1}{(x+1)^3} e^x dx$

$= 1 - 3\int_0^1 \frac{e^x}{x+1} dx + 3\int_0^1 \frac{e^x}{(x+1)^2}\, dx - \left[\left(\frac{-1}{2(x+1)^2} e^x\right)_0^1 - \int_0^1 \frac{-1}{2(x+1)^2} e^x dx\right]$

$= 1 - 3\,A + 3\left(-\frac{e}{2} + 1 + A\right) - \left[\left(\frac{-e}{8} + \frac{1}{2}\right) + \frac{1}{2}\left(\frac{-e}{2} + 1 + A\right)\right]$

$= 1 - 3\,A - \frac{3e}{2} + 3 + 3\,A + \frac{e}{8} - \frac{1}{2} + \frac{e}{4} - \frac{1}{2} - \frac{A}{2} = 3 - \frac{A}{2} - \frac{9e}{8}$

**Q.15 Answer:**

10

**Solution:**

$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$ **...(1)**

$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{-2\sin^3 4x}} dx$

$I = \int_0^{10\pi} \frac{e^{2\sin^3 4x}(\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{2\sin^3 4x}} dx ...(2)$

**Add eq. (1) + eq. (2)** $2I = \int_0^{10\pi} \underbrace{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}_{f(x)} dx$

$f(x)$ **repeats after** $2\pi$

$2I = 5\int_0^{2\pi} f(x)dx$, **again** $f(2a - x) = f(x)$

$2I = 10\int_0^{\pi} f(x)dx$, **again** $f(\pi - x) = f(x)$

$2I = 20\int_0^{\pi/2} f(x)dx \Rightarrow I = 10\int_0^{\pi/2} f(x)dx$

$\therefore \quad I = 10\int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x dx$ **...(3)**

**Hence,** $k = 10$

**Q.16 Answer:**

5

**Solution:**

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$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$ ...(1)

$$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{-2\sin^3 4x}} dx$$

$$I = \int_0^{10\pi} \frac{e^{2\sin^3 4x}(\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1 + e^{2\sin^3 4x}} dx ...(2)$$

**Add eq. (1) + eq. (2)** $2\mathrm{I} = \int_0^{10\pi} \underbrace{\cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{x}}_{\mathrm{f(x)}} \mathrm{dx}$

$\mathrm{f(x)}$ **repeats after** $2\pi$

$2\mathrm{I} = 5\int_0^{2\pi} \mathrm{f(x)dx}$, **again** $\mathrm{f(2a - x) = f(x)}$

$2\mathrm{I} = 10\int_0^{\pi} \mathrm{f(x)dx}$, **again** $\mathrm{f(\pi - x) = f(x)}$

$2I = 20\int_0^{\pi/2} f(x)dx \Rightarrow I = 10\int_0^{\pi/2} f(x)dx$

$\therefore \quad \mathrm{I} = 10\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{xdx}$ **...(3)**

$I = 10\int_0^{\pi/2} \cos(3\pi - 6x)\cos\left(\frac{7\pi}{2} - 7x\right)\cos(4\pi - 8x)\cos\left(\frac{9\pi}{2} - 9x\right) dx$

$I = 10\int_0^{\pi/2} \cos 6x \cdot \sin 7x \cos 8x \cdot \sin 9x dx$ **...(4)**

**Eq. (3) + Eq. (4)**

$2\mathrm{I} = 10\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 8\mathrm{x}[\cos 7\mathrm{x} \cdot \cos 9\mathrm{x} + \sin 7\mathrm{x} \cdot \sin 9\mathrm{x}]\mathrm{dx}$

$2I = 10\int_0^{\pi/2} \cos 6x \cdot \cos 8x \cdot \cos 2x dx$

$I = 5\int_0^{\pi/2} \cos 6x \cdot \cos 8x \cdot \cos 2x dx$

**Hence,** $c = 5$

**Q.17 Answer:**

$\frac{5\pi}{8}$

**Solution:**

$\mathrm{I} = \int_0^{10\pi} \frac{\cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{x}}{1+\mathrm{e}^{2\sin^3 4\mathrm{x}}} \mathrm{dx}$

$\mathrm{I} = \int_0^{10\pi} \frac{\cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{x}}{1+\mathrm{e}^{-2\sin^3 4\mathrm{x}}} \mathrm{dx}$

$\mathrm{I} = \int_0^{10\pi} \frac{\mathrm{e}^{2\sin^3 4\mathrm{x}}(\cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{x}}{1+\mathrm{e}^{2\sin^3 4\mathrm{x}}} \mathrm{dx}$

$2I = \int_0^{10\pi} \underbrace{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}_{f(x)} dx$

$2\mathrm{I} = \int_0^{10\pi} \underbrace{\cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{x}}_{\mathrm{f(x)}} \mathrm{dx}$

$\mathrm{f(x)}$ **repeats after** $2\pi$

$2\mathrm{I} = 5\int_0^{2\pi} \mathrm{f(x)dx}$, **again** $\mathrm{f(2a - x) = f(x)}$

$2\mathrm{I} = 10\int_0^{\pi} \mathrm{f(x)dx}$, **again** $\mathrm{f(\pi - x) = f(x)}$

$2\mathrm{I} = 20\int_0^{\pi/2} \mathrm{f(x)dx} \Rightarrow \mathrm{I} = 10\int_0^{\pi/2} \mathrm{f(x)dx}$

$\mathrm{I} = 10\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 7\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 9\mathrm{xdx} \quad \ldots\ldots\ldots(3)$

$\mathrm{I} = 10\int_0^{\pi/2} \cos(3\pi - 6\mathrm{x})\cos\left(\frac{7\pi}{2} - 7\mathrm{x}\right)\cos(4\pi - 8\mathrm{x})\cos\left(\frac{9\pi}{2} - 9\mathrm{x}\right) \mathrm{dx}$

$\mathrm{I} = 10\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \sin 7\mathrm{x} \cos 8\mathrm{x} \cdot \sin 9\mathrm{xdx} \ldots\ldots\ldots\ldots(4)$

**Eq.** $(3)+$ **Eq. (4)**

$2\mathrm{I} = 10\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 8\mathrm{x}[\cos 7\mathrm{x} \cdot \cos 9\mathrm{x} + \sin 7\mathrm{x} \cdot \sin 9\mathrm{x}]\mathrm{dx}$

$2\mathrm{I} = 10\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 2\mathrm{xdx}$

$\mathrm{I} = 5\int_0^{\pi/2} \cos 6\mathrm{x} \cdot \cos 8\mathrm{x} \cdot \cos 2\mathrm{xdx}$

$\mathrm{I} = 5\int_0^{\pi/2} \frac{\cos 8\mathrm{x}}{2}(\cos 8\mathrm{x} + \cos 4\mathrm{x})\mathrm{dx} = \frac{5}{2}\int_0^{\pi/2} \cos 8\mathrm{x}(\cos 4\mathrm{x} + \cos 2\mathrm{x})\mathrm{dx}$

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$= \frac{5}{2}\int_0^{\pi/2}\cos^2 8x dx + \frac{5}{2}\int_0^{\pi/2}\cos 4x\cos 2x dx = \frac{1}{8}\cdot\frac{5}{2}\int_0^{4\pi}\cos^2 t dt \quad (8x = t) = \frac{5}{4}\int_0^{\pi}\cos^2 t dt$

$I = \frac{5}{2}\int_0^{\pi/2}\cos^2 t dt = \frac{5}{2}\int_0^{\pi/2}\sin^2 t dt$

$2I = \frac{5}{2}\int_0^{\pi/2} dt \Rightarrow I = \frac{5}{4}\cdot\frac{\pi}{2} \Rightarrow I = \frac{5\pi}{8}$

**Q.18 Answer:**

A-s ; B-s ; C-q,r,s ; D-s

**Solution:**

**(A) - s, (B) - s, (C) - qrs, (D) - s**

**(A)** $F(x) = \int \frac{x+\sin x}{1+\cos x}dx = \int \left(x\frac{1}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}\right)dx = x\tan\frac{x}{2} + c$

**Since** $0 = F(0)$

$\therefore \quad C = 0$ **and** $F(\pi/2) = \pi/2$

**(B)** $F(x) = \int e^{\sin^{-1}x}\left(1 - \frac{x}{\sqrt{1-x^2}}\right)dx = \int e^{\sin^{-1}x}\left(\frac{1}{\sqrt{1-x^2}}\sqrt{1-x^2} + \frac{-x}{\sqrt{1-x^2}}\right)dx$

$$F(x) = e^{\sin^{-1}x}\sqrt{1-x^2} + c$$
$$F(0) = 1 + c \Rightarrow c = 0$$
$$F(1/2) = e^{\pi/6}\cdot\frac{\sqrt{3}}{2} = \frac{k\sqrt{3}}{\pi}e^{\pi/6} \therefore k = \frac{\pi}{2}$$

**(C)** $\int \frac{dx}{(x^2+1)(x^2+9)} = \frac{1}{8}\int\left(\frac{1}{x^2+1} - \frac{1}{x^2+9}\right)dx$

$\frac{1}{8}\left[\tan^{-1}x - \frac{1}{3}\tan^{-1}\frac{x}{3}\right] + c$

$F(0) = c \Rightarrow c = 0$

$\therefore \quad \frac{1}{8}\left(\frac{\pi}{3} - \frac{1}{3}\cdot\frac{\pi}{6}\right) = \frac{5\pi}{144} = \frac{5k}{36} \quad \therefore k = \frac{\pi}{4}$

**(D)** $F(x) = \int \frac{\sqrt{\tan x}}{\sin x\cos x}dx = \int(\tan x)^{-1/2}\sec^2 x dx = 2\sqrt{\tan x} + c$

$F(0) = c$

$\therefore c = 0$

$\therefore F(\pi/4) = 2 = \frac{2k}{\pi} \quad \therefore k = \pi.$

**Q.19 Answer:**

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**Solution:**

$\int_0^{2\pi}\ln\left(\sec^2 x\right)dx = 2\int_0^{\pi}\ln\left(\sec^2 x\right)dx$

$= 4\int_0^{\pi/2}\ln\left(\sec^2 x\right)dx = 8\int_0^{\pi/2}\ln(\sec x)dx = -8\int_0^{\pi/2}\ln(\cos x)dx = -8\left(-\frac{\pi}{2}\ln 2\right)$

$= 4\pi\ln 2 = 2k^2\sin^{-1}\left(\frac{k}{2}\right)\ln k$

$\therefore \quad k = 2$

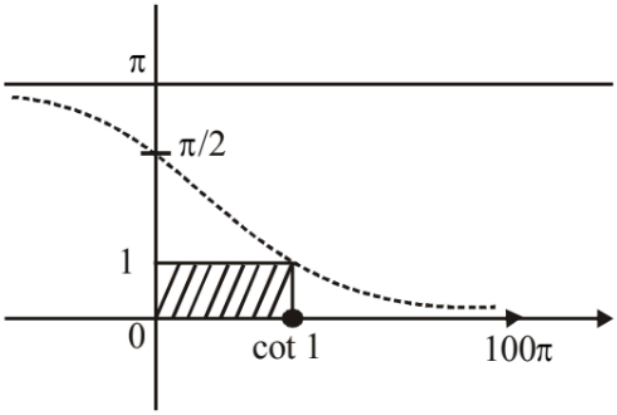
$\therefore \quad [k] = 2$

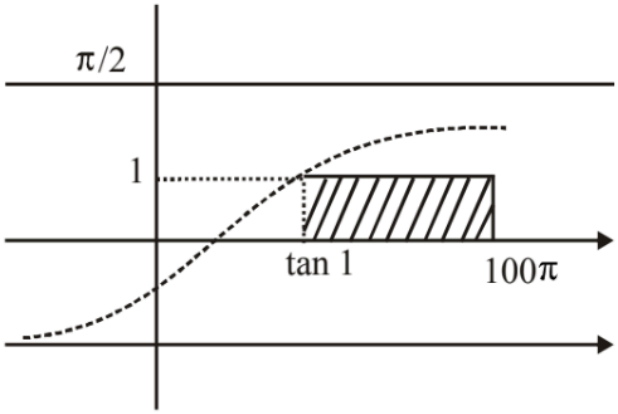
**Q.20 Answer:**

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**Solution:**

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$$\int_0^{100\pi} \left(\left[\cot^{-1} x\right] + \left[\tan^{-1} x\right]\right) dx = \cot 1 + (100\pi - \tan 1) = 100\pi + \frac{1 - \tan^2 1}{\tan 1} = 100\pi + 2\cot 2$$

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